



**Madhav Gupta**  
**Electrical Engineering**  
**Indian Institute of Technology Bombay**  
**Specialization: Communication & Signal Processing**

**21D070043**  
**Dual Degree (B.Tech. + M.Tech.)**  
**Gender: Male**  
**DOB: 06/08/2003**

| Examination | University | Institute  | Year | CPI / % |
|-------------|------------|------------|------|---------|
| Graduation  | IIT Bombay | IIT Bombay | 2026 |         |

Pursuing Minor degree in **Artificial Intelligence and Data Science** offered by CMInDS, IIT Bombay

## SCHOLASTIC ACHIEVEMENTS

- Achieved **99.989 percentile** and All India Rank **164** in **JEE Main** examination amongst 1 million+ students [2021]
- Secured All India Rank **724** out of 0.15 million+ candidates who appeared for **JEE Advanced** examination [2021]
- Recipient of Kishore Vaigyanik Protsahan Yojana (**KVPY**) Fellowship organised by **IISc Bangalore** [2021]
- Recipient of the prestigious National Talent Search Examination (**NTSE**) scholarship, ranked **3rd** in Stage-1 [2019]
- Qualified for Regional Maths Olympiad(**RMO**) by securing a rank in **Top 1%ile** in Pre-RMO by HBCSE [2018]

## PUBLICATIONS

- Kavitha Viswanathan, **Madhav Gupta**, Amit Sethi et al., "**FANVID: A Benchmark for Face and License Plate Recognition in Low-Resolution Videos**", submitted to **NeurIPS 2025**(Datasets and Benchmarks Track, under review)

## PROFESSIONAL EXPERIENCE

### Full-Stack AI Intern | NuSummit (National Stock Exchange-IT)

[May '25 - Aug '25]

- Built a POC **RAG-based Q&A** system for financial documents using **Ollama** & **Qdrant**, handling PDFs of 10+ pages
- Integrated end-to-end pipeline using **FastAPI** with user auth and enhanced Angular UI for secure **PDF querying**
- Explored **RAGAS** framework to implement guardrails and enhance **context matching** for accurate response evaluation

### AI Research Intern | Sentry AI

[Jan '25 - May '25]

Awarded a **Letter of recommendation** for outstanding work on real-world computer vision problems

- Boosted loitering detection by **50%** via **contrastive learning** for background-invariant clustering across 50+ cameras
- Fine-tuned a Swin Transformer with **K-fold cross-validation** to segregate suitable images, achieving **83%** accuracy
- Slashed false positives by **80%** across **100+** cameras using height-vs-(X,Y) coordinate filtering via **2D linear regression**

### Generative AI Solutions Intern | Business Unit-CSW | Hilti Group

[Sep '24 - Jan '25]

Awarded a **Pre-Placement Interview** for full-time return offer

- Integrated **4** apps via APIs, SharePoint, & **Copilot Studio**, boosting info search efficiency by **3x** across platforms
- Implemented **API connectivity** with **topic triggers** to fetch relevant information and generate RAG-style answers
- Presented a successful **POC** to leadership, securing approval for future scaled product development for **10K+** users

## KEY PROJECTS

### Applied Quantitative Methods in Finance

[Jun '25 - Jul '25]

Summer of Quant | Quant Community, IITB

- Completed a 7-week program covering stochastic calculus, **financial derivatives**, and **time series forecasting**
- Designed and backtested **BTC** using daily **LONG, SHORT & EXIT** signals from an initial \$1000 capital

### Limestone Data Challenge | Algorithmic Trading Hackathon

[Mar '23 - Apr '23]

Organised by **Finance & Analytics Club**, IIT Bombay in association with **Tower Research Capital**

- Classified 100 stocks into **4** sectors using **hierarchical clustering** and correlation analysis based on their returns data
- Predicted **index returns** using sectoral data, employing **regression** for classification and performance analysis metrics
- Implemented an optimized portfolio allocation strategy with a **Sharpe ratio** over **1.72**, incorporating risk preferences

### Statistical and Predictive Analysis | Course Project

[Jan '25 - Apr '25]

DH302 - Introduction to Public Health Informatics | Guide: Prof. Saket Choudhary

- Modeled **2M+** NEET-UG'24 scores with **AIC-scored log-normal** fit; compared with '23 baseline to flag leak anomalies
- Applied **PCA** to 18 amino-acid features from 612 foods & used **hypothesis testing** to validate nutrient-based clusters
- Built **gamma GLM** using AIC on **5+** states to predict monthly conceptions from temperature and humidity trends

### Credit Card Fraud Detection | Convolve 3.0

[Jan '25 - Feb '25]

AI/ML Hackathon organized by Top 8 IITs and presented by IDFC First Bank

- Addressed issue of **class imbalance** via **SMOTE**-based data pipeline & filtered high impact features using **KS statistic**
- Achieved an ROC-AUC of **0.85** with **XGBoost classifier**, which outperformed Random Forest and Logistic Regression

## TECHNICAL & RESEARCH EXPERIENCE

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### IIT Bombay Racing - Driverless

Part of a 70+ member team designing driverless electric race cars for Formula Student competitions in Germany/UK  
**Formula Student AI 2023 | Representing the Team at Silverstone** [May '23 - Jul '23]

- Part of the **First ever contingent** of the only Indian team to participate in FS-AI 2023 at the Silverstone Circuit
- Engineered **CAN**(Controller Area Network) interface between **AI** unit and vehicle control unit of **Driverless Car**
- Delivered a **Business Plan Presentation** for an advanced **Level 4 autonomous system** to international judges

**Robust Visual Attribution via Submodular Optimization for OOD Data** [Jan '25 - Present]  
*Thesis Project | Guide: Prof. Ganesh Ramakrishnan*

- Built a submodular selection and **perturbation-based uncertainty** framework for enhanced vision model interpretability
- Achieved higher Insertion AUC(**0.91** on synthetic OOD) over prior in-distribution submodular models with **ResNet-101**
- Drafting a research paper for **IEEE WACV 2025** on robust visual attribution for **Out-of-Distribution** scenarios

## OTHER PROJECTS

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**Stock Price Prediction using LSTM | Course Project** [Aug '23 - Sep '23]  
*EE782 - Advanced Machine Learning | Guide: Prof. Amit Sethi*

- Analyzed minute-by-minute stock data using **candlestick plots** & employed linear extrapolation for data cleaning
- Engineered PyTorch LSTM models and crafted an **adaptable data loader** for dynamic stock price prediction
- Designed a **trading strategy class** incorporating **EMA indicator** to execute stock trades based on model predictions

**Understanding Trading of Currencies and Commodities in Stock Market** [Jun '22 - Aug '22]  
*Financial Research Program - FinSearch | Finance Club, IIT Bombay*

- Led a team of 4 through a financial research project & stood among **top 15** from more than **250** participating teams
- Proficiently studied and analyzed **forex & commodity market**, assessing the impact of key factors such as **inflation, politics, economic growth, supply & demand, interest rates, government debts, and economic stability**
- Tested models for commodity & forex trading, then **backtested** them against some to check their correctness

**AI-Powered Tender Discovery and Retrieval System | Industrial Project** [Jan '25 - Mar '25]  
*Convolution Engineering Consultancy | Transforming tender management using AI*

- Built semantic search using **Pinecone**, E5 embeddings and **Reranker** for vector-based ranking across **10K+** tenders
- Implemented **full lifecycle automation**, handling **1K+** new tenders daily, expiry-based deletion and real-time downloads
- Automated **tender scraping** across government portals using **Selenium** generating JSONs with key metadata entities

**AI Powered Resume Assistant | Course Project** [Oct '23 - Nov '23]  
*EE782 - Advanced Machine Learning | Guide: Prof. Amit Sethi*

- Fine-tuned a **Mistral-7B** model on the scraped IITB resume repository to capture **domain-specific** writing styles
- Utilized **4-bit quantized** and **sharded models** for parallel inference to optimize GPU usage and fine-tune efficiently

## TECHNICAL PROFICIENCY AND RELEVANT COURSES

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- **Programming Languages** : Python, JavaScript, TypeScript, C++, C#, SQL, bash
- **Frameworks** : Git, Linux, Angular, .NET Core, Specflow, Selenium, MongoDB, CI/CD pipelines, Copilot Studio
- **Libraries** : NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, SciPy, Scikit-Learn, OpenCV, NLTK
- **Key Courses** : Probability and Random Processes, Data Structure and Algorithms, Programming for Data Science, Advanced Machine Learning, Optimization in Machine Learning, Markov Chains

## POSITION OF RESPONSIBILITY

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**Interview Coordinator | Placement Office, IIT Bombay** [Aug '22 - Dec '22]

- Coordinated with a team of **250+** members to interview 2000+ students during the **placement season** of 2022-23
- Assisted in conducting Interviews for **20+** firms, handling **student queries**, and maintaining interview records

**Event Coordinator | Impulse, EESA's Annual Fest** [Mar '23]

- Acted as a **point of contact** (POC) for top management from companies like **Samsung, Mercedes & Reliance Jio**
- Got a chance to interact with them, fostering **valuable connections** and enhancing the **reputation** of our institution

## EXTRA CURRICULAR ACTIVITIES

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- Secured **2nd** position in **QuantHive's Algoswarm, Algo-Trading Hackathon**,organised by Finance Club [Aug '23]
- Attained **Gold Level** in **WorldQuant BRAIN** for contributions in quantitative modeling and strategy [Oct '23]
- Contributed as a member of the **10-person team** for **DevRev Agent challenge** at **Inter IIT Tech Meet 12.0**[Dec '23]
- Mentored **10+** participants in **WiDS Bootcamp** on developing a **Sudoku Solver** using **Computer Vision** [Dec '23]
- Achieved **3rd place** in the Electrical Engineering department-wide **table tennis** tournament [Aug '24]